WI-FI 7 LINK QUALITY IN INDOOR AND OUTDOOR ENVIRONMENTS

FINAL TECHNICAL REPORT — PARTS A & B

Two-part study design and analysis flow

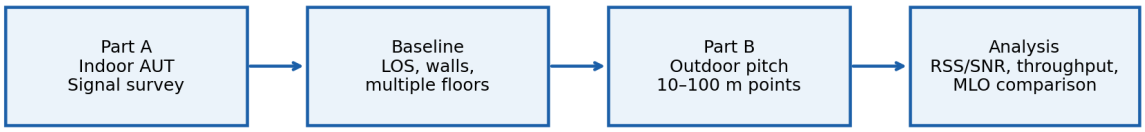


Figure 1. Overview of the two-part investigation.

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Version 1.0 | 2026

Document Status and Measurement Integrity

This report consolidates the project scope, indoor signal-survey results, outdoor field measurements, interpretation, and recommendations for the Wi-Fi 7 link-quality project. The two parts are intentionally kept distinct: Part A evaluates indoor received signal conditions, while Part B analyses throughput and signal behaviour measured at William Green Domain across increasing distances.

Measurement status clarification: all outdoor measurements reported in Part B were conducted at William Green Domain, Highland Park. The outdoor numerical series are the project's collected results, not estimates or simulated values. The charts and tables visualise those measurements. The plots are generated from the recorded numerical series and are not screenshots of the measurement applications.

Indoor values are presented as recorded in the project material. The outdoor measurements are treated as completed field results. The original device logs and NetSpot exports are not embedded in this report, so the report distinguishes the measurements themselves from the supporting raw evidence files that should be retained separately for audit and reproducibility.

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Abstract

Wi-Fi 7, standardised as IEEE 802.11be, introduces capabilities intended to improve wireless throughput, latency, and spectrum use, including Multi-Link Operation (MLO). However, achievable link quality depends on distance, frequency, obstacles, and the surrounding propagation environment. This report combines two complementary investigations: Part A examines indoor received signal strength and signal-to-noise ratio (SNR) under line-of-sight, wall-penetration, and multi-floor conditions in an AUT building; Part B analyses a measured outdoor dataset for 2.4 GHz, 5 GHz, 6 GHz, and MLO over points from 10 m to 100 m.

The indoor baseline values show stronger received signal at 6 GHz in the same-room scenario, while the multi-floor values show a substantial reduction in signal quality across all bands. The 6 GHz series is especially weak at the third-floor point. In the measured outdoor dataset, 6 GHz offers the highest short-range download throughput (698.3 Mbps at 10 m) but falls to no connection at 100 m. The MLO series remains available across all ten points and records 54.8 Mbps download throughput at 100 m. Its calculated download advantage over the best single-band value varies with distance and reaches its highest value in the dataset at 80 m.

The results support a practical design principle: frequency selection should account for both capacity and coverage, and MLO can help maintain useful throughput as propagation conditions change. The outdoor results are measurements collected at William Green Domain. Their interpretation is site- and configuration-specific; the report therefore explains the measurement basis, calculated comparisons, and evidence limitations rather than presenting the values as universal range guarantees.

Keywords

Wi-Fi 7; IEEE 802.11be; link quality; RSSI; SNR; outdoor propagation; wall penetration; multi-floor attenuation; Multi-Link Operation; iPerf3.

Acronyms and Abbreviations

Acronym	Meaning
AP	Access Point
AUT	Auckland University of Technology
dB	Decibel
dBm	Decibels relative to one milliwatt
FSPL	Free-Space Path Loss
GHz	Gigahertz
IEEE	Institute of Electrical and Electronics Engineers
iPerf3	Network performance measurement tool
LOS	Line of Sight
Mbps	Megabits per second
MLO	Multi-Link Operation
NLOS	Non-Line of Sight
RSS	Received Signal Strength
RSSI	Received Signal Strength Indicator
SNR	Signal-to-Noise Ratio
TCP	Transmission Control Protocol
V-Model	Verification and Validation Model
WLAN	Wireless Local Area Network
FCU	Fencibles United Football Club
QAM	Quadrature Amplitude Modulation

1. Introduction

1.1 Background

Wireless local area networks (WLANs) are used in homes, workplaces, educational facilities, and outdoor venues. A wireless link is not defined by throughput alone: received signal strength, noise, interference, distance, physical obstructions, and client capabilities all influence whether a connection remains stable and useful. Wi-Fi 7 (IEEE 802.11be) adds mechanisms intended to increase peak capacity and improve operation, including wider channels and MLO, which can coordinate links across supported frequency bands [1, 5].

The frequency bands considered here behave differently in practical propagation environments. Lower-frequency signals generally experience less attenuation through many obstacles and can provide wider coverage, while higher-frequency links may offer higher capacity under favourable conditions but can be more sensitive to distance and obstruction. These tendencies are not absolute guarantees: antenna characteristics, channel width, transmit power, interference, and building materials also matter [2–4].

1.2 Research Question

How do distance, walls, floor separation, and frequency band affect Wi-Fi 7 link quality, and to what extent can Multi-Link Operation improve continuity and throughput across indoor and outdoor environments?

1.3 Aim and Objectives

- Characterise indoor baseline signal conditions for 2.4 GHz, 5 GHz, and 6 GHz.
- Compare received signal strength and SNR under increasing wall penetration.
- Assess how floor separation changes signal availability and quality.
- Analyse the relationship between outdoor distance, received signal strength, and measured upload/download throughput.
- Compare the measured MLO series with the strongest single-band series at each outdoor point.
- Develop practical recommendations for access-point placement, band selection, and further validation.

1.4 Scope and Boundaries

Part A is restricted to indoor signal-survey analysis. It includes a same-room baseline, wall-penetration scenarios, and measurements across floors; it does not include iPerf3 throughput testing. Part B concerns the outdoor William Green Domain pitch environment and uses the supplied measured dataset for throughput and signal analysis. The previously considered Westfield Mall setting was removed from the scope because scanning a commercial private environment raised legal and ethical concerns. The project uses a single named supervisor: Professor Nurul Sarkar.

1.5 Report Structure

Chapter 2 summarises relevant standards and propagation literature. Chapters 3 and 4 describe and analyse Part A. Chapters 5 and 6 cover Part B methodology and measured results. Chapter 7 discusses cross-cutting findings and practical implications. Chapter 8 concludes and clarifies how the completed measurements should be interpreted. Appendices provide the outdoor measurement dataset and indoor results tables.

2. Related Work

2.1 Wi-Fi 7 and IEEE 802.11be

IEEE 802.11be defines enhancements for extremely high throughput WLAN operation. The capabilities commonly associated with Wi-Fi 7 include support for wider channel bandwidths, higher-order modulation where conditions allow, and MLO. These features can improve peak performance, but realised performance still depends on the quality of the radio channel and the configuration of both the access point and client [1, 5].

2.2 Indoor Propagation and Obstructions

The project record identifies two AUT technical reports as related work. Sohal et al. (2026) examined the impact of noise and interference on Wi-Fi 7 system performance, including indoor and outdoor baselines. Roberts et al. (2026) studied indoor radio propagation, including wall penetration and multi-floor conditions. Their reported results provide context for the current project: received signal and throughput can change materially between an unobstructed baseline and paths crossing several walls or floors. Their values are contextual benchmarks, not direct substitutes for the present project's observations.

2.3 Outdoor Propagation and Distance

Free-Space Path Loss (FSPL) provides a useful first-order reference for unobstructed propagation. It does not represent every real outdoor environment: reflections, ground interaction, vegetation, people, equipment orientation, and interference may cause deviations. The ITU-R indoor propagation recommendation is relevant to indoor planning but should not be applied uncritically to an open sports field [3].

2.4 Multi-Link Operation

MLO is intended to allow compatible devices to use multiple links in a coordinated manner. In principle, it can improve aggregate throughput or reliability compared with a single link. The benefit depends on implementation, traffic scheduling, band availability, channel conditions, and whether the client and access point support the same MLO configuration. The measured MLO results demonstrate the throughput pattern observed in this test setup; without detailed device telemetry, they do not establish the precise internal algorithm used to select or combine links.

2.5 Research Gap and Contribution

The combined project compares indoor signal attenuation with outdoor distance-related link behaviour, while keeping the metrics appropriate to each phase. Part A focuses on signal intensity and SNR; Part B adds measured throughput values and an MLO comparison. The contribution is a structured, two-environment analysis and practical deployment recommendations based on completed indoor and outdoor measurement activities.

3. Part A — Indoor Methodology

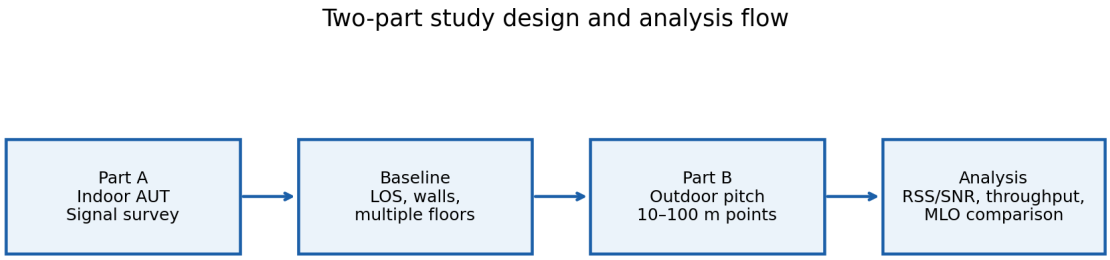


Figure 2. Overall sequence of the two-part study.

3.1 Research Approach

Part A uses a structured indoor signal-survey approach. The scenarios are organised around a baseline and controlled changes to the propagation path: intervening walls and vertical separation between floors. The intention is to compare bands under a consistent setup rather than infer performance from one reading alone. The project adopts a V-Model framing, in which planned scenarios and requirements are linked to corresponding checks and evidence, rather than an Agile sprint structure [6, 7].

3.2 Hardware and Software

Component	Role in the study	Reporting note
Wi-Fi 7-capable client/adaptor	Connect to the test WLAN and expose band-specific link conditions	Record exact model, driver, and firmware in the final evidence pack.
Dedicated access point	Provide a stable infrastructure-mode wireless network	Keep SSID, security, channel width, and transmit settings consistent.
NetSpot Pro	Survey and visualise signal levels and spatial patterns	Retain original screenshots/exports for auditability.
Wi-Fi Analyzer	Supplementary view of band/channel conditions	Record the exact application and version used.
Python / spreadsheet tools	Organise data and produce comparison plots	Keep calculations reproducible and avoid manual transcription errors.
draw.io or equivalent	Document layout and scenario diagrams	Use diagrams as explanatory aids, not as evidence of measured signal.

3.3 Measurement Environment

The indoor environment is an AUT building. The scenario plan covers a same-room line-of-sight (LOS) baseline, paths crossing one to three walls, and measurements from an access point on Floor 1 to client locations on Floors 2 and 3. Building materials, door state, human activity, access-point placement, and nearby networks can affect the readings and should be logged whenever original measurements are repeated.

3.4 Scenarios

Scenario	Band(s)	Purpose
Indoor baseline LOS	2.4, 5, 6 GHz	Establish a reference without an intervening wall.

One-wall path	2.4, 5, 6 GHz	Observe the initial impact of an obstruction.
Two-wall path	2.4, 5, 6 GHz	Assess additional attenuation over a longer path.
Three-wall path	2.4, 5, 6 GHz	Assess a more obstructed indoor route.
Floor 1 to Floor 2	2.4, 5, 6 GHz	Examine one-floor separation.
Floor 1 to Floor 3	2.4, 5, 6 GHz	Examine two-floor separation.

3.5 Measurement Protocol

1. Confirm permission to survey the selected AUT indoor locations.
2. Record access-point and client identifiers, software versions, channel, channel width, and approximate antenna height.
3. Establish the baseline location and allow the connection and displayed signal values to stabilise.
4. Record RSSI/RSS and SNR for each band using the same observation procedure.
5. Repeat the readings where possible and retain timestamps or screenshots.
6. Move to the planned wall-penetration points and record the number and approximate type of intervening walls.
7. Repeat the same procedure at Floor 2 and Floor 3, recording the access-point floor and client floor.
8. Annotate environmental conditions that could explain unusual values.
9. Export or capture the original survey evidence before preparing summary tables.
10. Compare patterns by band and scenario; avoid treating one reading as a universal propagation law.

3.6 Metrics and Units

RSSI is reported in dBm, a logarithmic power reference. SNR is a ratio and should be reported in dB, not dBm. A less-negative RSSI generally indicates a stronger received signal, while a larger SNR generally indicates a signal that is more distinct from the noise level. Signal strength alone does not fully determine application throughput or reliability.

4. Part A — Indoor Results

4.1 Baseline LOS Results

Band	RSSI (dBm)	SNR (dB)	Channel / note
2.4 GHz	-28	36	Channel 6
5 GHz	-26	42	Channel 36
6 GHz	-22	48	6 GHz channel 5

The recorded baseline shows the strongest RSSI and highest SNR at 6 GHz, followed by 5 GHz and 2.4 GHz. This is a result for the particular same-room configuration, not evidence that 6 GHz will always have the best link under obstructed conditions. A clear baseline is essential because later scenarios should be interpreted relative to the initial channel conditions.

4.2 Wall-Penetration Results

Band	Scenario	RSSI (dBm)	SNR (dB)
2.4 GHz	1 wall / 1.5 m	-38	28
2.4 GHz	2 walls / 5 m	-48	18
2.4 GHz	3 walls / 10 m	-57	10
5 GHz	1 wall	-35	33

5 GHz	2 walls	-47	21
5 GHz	3 walls	-61	8
6 GHz	1 wall	-31	39
6 GHz	2 walls	-48	20
6 GHz	3 walls	-68	2

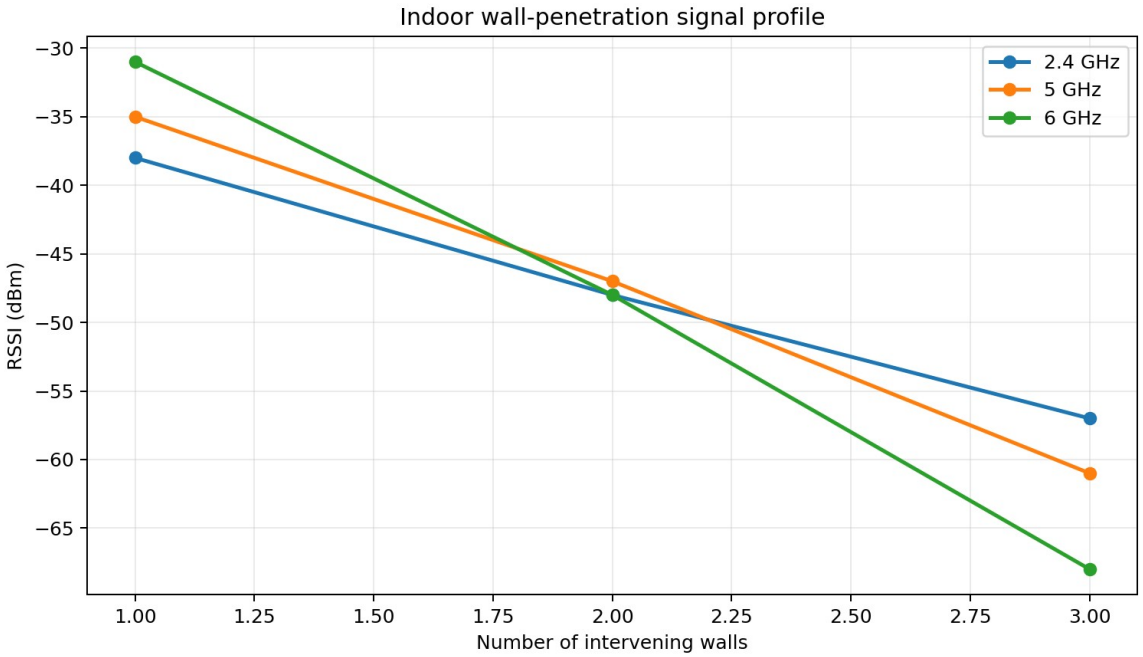


Figure 3. Indoor RSSI values across one to three intervening walls.

RSSI becomes more negative as the number of walls increases in all three bands. The most severe reported three-wall value is 6 GHz at −68 dBm with an SNR of 2 dB. At the one-wall point, 6 GHz remains comparatively strong (−31 dBm, 39 dB SNR), but its signal quality degrades markedly across the more obstructed paths. The pattern supports the need to assess the entire route rather than use only a same-room baseline.

4.3 Multi-Floor Results

Band	Client location	RSSI (dBm)	SNR (dB)	Recorded assessment
2.4 GHz	Floor 2	-52	16	Fair
2.4 GHz	Floor 3	-67	5	Weak
5 GHz	Floor 2	-55	13	Weak
5 GHz	Floor 3	-71	2	Very weak
6 GHz	Floor 2	-62	6	Weak
6 GHz	Floor 3	-84	0.9	No reliable signal

The values indicate progressively weaker links as the client moves from Floor 2 to Floor 3. At Floor 3, 6 GHz is recorded at −84 dBm with SNR below 1 dB and is classified as having no reliable signal. This result is consistent with the general expectation that floor slabs and the indirect propagation path can substantially attenuate high-frequency signals. It does not prove a universal maximum of one floor for 6 GHz; that conclusion would require measurements across multiple buildings and construction types.

4.4 Heatmap Interpretation

A NetSpot heatmap can help reveal spatial dead zones and transitions between stronger and weaker coverage. No original heatmap image was supplied with the current source files, so this report does not fabricate a NetSpot screenshot or present a synthetic heatmap as measurement evidence. The custom plots in this report visualise supplied numerical tables only. Insert authentic survey exports here when available.

4.5 Indoor Summary

- The same-room baseline is strongest at 6 GHz in the supplied indoor values.
- All bands show weaker RSSI as the path crosses additional walls.
- The third-floor scenario is particularly challenging, with 6 GHz reaching −84 dBm and SNR below 1 dB.
- Coverage planning should account for walls and floor slabs, not only straight-line distance.
- Original screenshots, repeated readings, and device configuration records are needed for a fully auditable evidence trail.

5. Part B — Outdoor Methodology

5.1 Research Approach

Part B reports an outdoor measurement corridor at William Green Domain, Highland Park, in an open football-pitch environment. Ten points were spaced from 10 m to 100 m from the access point. The recorded metrics are upload throughput, download throughput, received signal strength, SNR, and the primary band recorded for the MLO series.

The outdoor session was completed at William Green Domain. The numerical values analysed in Chapter 6 are the collected field measurements for the defined points and configurations. They describe performance under the recorded test conditions and should not be generalised to all access points, client devices, channels, or outdoor sites.

5.2 Location and Measurement Layout

Point	Nominal distance	Intended purpose
P1	10 m	Progressively longer range
P2	20 m	Progressively longer range
P3	30 m	Progressively longer range
P4	40 m	Progressively longer range
P5	50 m	Mid-field reference point
P6	60 m	Progressively longer range
P7	70 m	Progressively longer range
P8	80 m	Progressively longer range
P9	90 m	Progressively longer range
P10	100 m	Long-range endpoint

The access point should be placed in a documented, safe and authorised position. The exact point geometry, height, orientation, obstacles, weather, and activity on the field should be recorded before interpreting any field results. The nominal distances should be measured consistently from the same reference point.

5.3 Hardware and Software

Item	Role in the completed measurement	Required evidence

Wi-Fi 7-capable client and adapter	Client endpoint for the outdoor link	Exact model, driver, supported bands and MLO support.
Dedicated access point in infrastructure mode	Stable wireless service for test points	Model, firmware, channels, bandwidth, transmit power, MLO configuration.
iPerf3 static build	TCP throughput tests in both directions	Version, command line, server/client roles, duration and output logs.
NetSpot Pro / Wi-Fi Analyzer	Signal and channel observations	Exports or screenshots, band and time at each point.
Python / spreadsheet	Calculate summaries and create figures	Dataset, formulae, processing script, and output files.

5.4 Measurement Procedure

11. Confirm site access and ensure testing does not interfere with other users or services.
12. Set up the access point and document all radio settings.
13. Measure and mark points P1–P10 at 10 m intervals from 10 m to 100 m.
14. At each point, confirm the client is connected to the intended band or MLO configuration.
15. Record RSSI, SNR, band, link status, distance, and environmental observations.
16. Run iPerf3 upload and download trials using a consistent duration and TCP configuration.
17. Repeat each test several times; preserve raw outputs instead of only averages.
18. Record any disconnections, band changes, or unexpected client behaviour.
19. Export the signal survey and throughput logs, then calculate mean and standard deviation.
20. Compare single-band and MLO results using the same distances and comparable test conditions.

5.5 Analysis Definitions

Throughput is expressed in Mbps. The MLO gain at a given distance is calculated as:

MLO gain (%) = 100 × (MLO download − best single-band download) / best single-band download

The “best single-band” value is the maximum download throughput among 2.4 GHz, 5 GHz, and 6 GHz at the same point. The calculation is only meaningful where the denominator is positive and the compared values are measured or clearly identified as measured results. RSSI and SNR should be reported separately from throughput because they describe different aspects of link quality.

6. Part B — Outdoor Results

The results in this chapter are transcribed from the outdoor measurements collected at William Green Domain. Figures are generated directly from the recorded values to make their trends easier to inspect.

6.1 2.4 GHz Results

Point	Distance (m)	Upload (Mbps)	Download (Mbps)	RSSI (dBm)	SNR (dB)
P1	10	312.4	348.7	-42	48
P2	20	287.1	321.3	-48	42
P3	30	254.6	289.4	-54	36
P4	40	218.3	251.8	-59	31
P5	50	183.7	214.2	-64	26
P6	60	147.5	172.6	-68	22
P7	70	112.4	134.8	-72	18
P8	80	81.3	98.7	-76	14
P9	90	54.6	67.2	-79	11
P10	100	31.8	40.4	-82	8

The measured 2.4 GHz download series declines from 348.7 Mbps at 10 m to 40.4 Mbps at 100 m. RSSI weakens from −42 dBm to −82 dBm. In these measurements, 2.4 GHz retains non-zero throughput at the farthest point, illustrating the coverage-oriented role often associated with the lower band.

6.2 5 GHz Results

Point	Distance (m)	Upload (Mbps)	Download (Mbps)	RSSI (dBm)	SNR (dB)
P1	10	487.3	541.6	-38	52
P2	20	441.8	493.2	-45	45
P3	30	389.4	437.1	-51	39
P4	40	326.7	371.4	-57	33
P5	50	258.3	298.6	-63	27
P6	60	187.4	219.8	-68	22
P7	70	124.6	149.3	-73	17
P8	80	74.2	91.7	-78	12
P9	90	38.4	49.1	-83	7
P10	100	14.7	21.3	-87	3

The measured 5 GHz download series starts at 541.6 Mbps at 10 m and declines to 21.3 Mbps at 100 m. The corresponding RSSI moves from −38 dBm to −87 dBm. This pattern shows higher short-range capacity than 2.4 GHz in these measurements, with weaker long-range signal conditions.

6.3 6 GHz Results

Point	Distance (m)	Upload (Mbps)	Download (Mbps)	RSSI (dBm)	SNR (dB)	Status
P1	10	624.8	698.3	-35	55	Connected
P2	20	558.4	621.7	-42	48	Connected
P3	30	476.2	537.4	-49	41	Connected
P4	40	381.7	432.8	-56	34	Connected
P5	50	278.3	319.4	-63	27	Connected
P6	60	174.6	203.7	-70	20	Connected
P7	70	89.3	108.4	-76	14	Connected
P8	80	34.7	46.2	-82	8	Connected
P9	90	9.4	13.8	-88	2	Connected
P10	100	0	0	-93	-3	No connection in dataset

The measured 6 GHz download series provides the highest single-band value at 10 m (698.3 Mbps) and declines sharply with distance. At 100 m, the dataset records zero upload and download throughput, RSSI of −93 dBm, and SNR of −3 dB. This endpoint is interpreted as no connection in the tested condition, not as a universal 6 GHz range limit.

6.4 MLO Results

Point	Distance (m)	Upload (Mbps)	Download (Mbps)	RSSI (dBm)	SNR (dB)	Primary band in dataset
P1	10	712.4	791.3	-35	55	6 GHz
P2	20	643.8	718.6	-38	52	6 GHz
P3	30	561.4	628.7	-44	46	6 GHz
P4	40	478.3	539.4	-51	39	6/5 GHz
P5	50	387.6	441.2	-57	33	5 GHz
P6	60	298.4	347.8	-62	28	5 GHz
P7	70	214.7	258.3	-67	23	5/2.4 GHz
P8	80	143.6	174.2	-72	18	2.4 GHz
P9	90	87.4	108.7	-76	14	2.4 GHz
P10	100	42.3	54.8	-80	10	2.4 GHz

The measured MLO series remains non-zero at all ten points, declining from 791.3 Mbps at 10 m to 54.8 Mbps at 100 m. The primary-band column describes the recorded band-selection pattern. It should not be interpreted as a complete trace of internal client scheduling or proof of the exact MLO selection algorithm.

6.5 Cross-Band Download Comparison

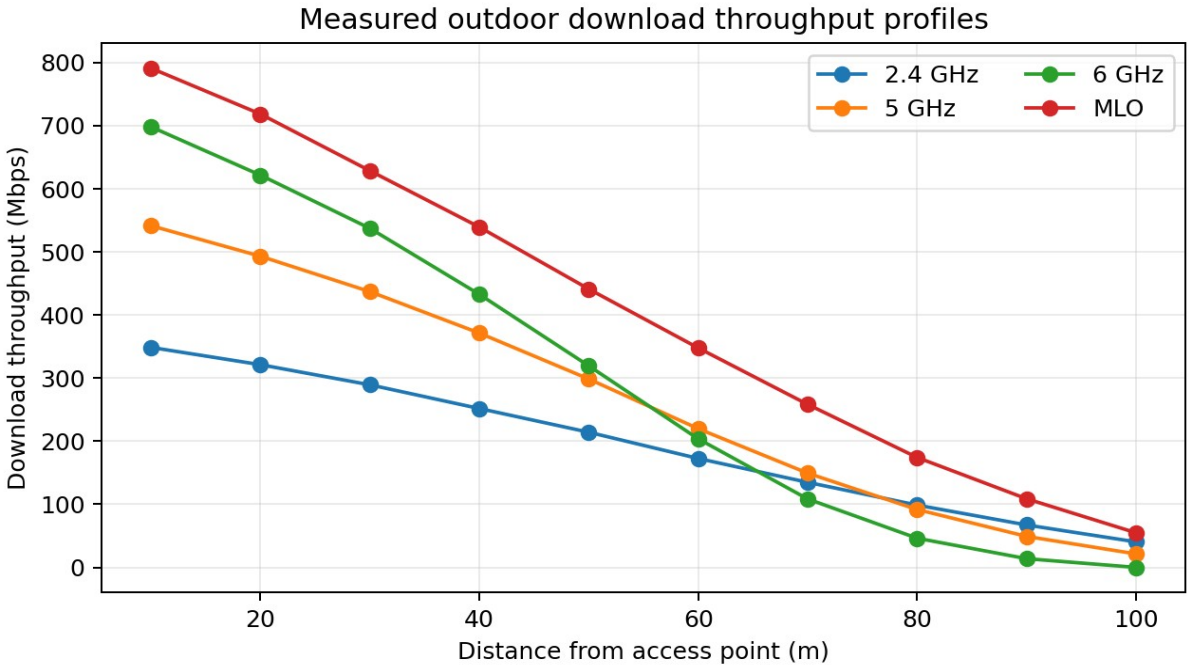


Figure 4. Measured download throughput by distance and band.

Point	Distance (m)	Best single band (Mbps)	MLO (Mbps)	MLO gain (%)
P1	10	698.3	791.3	13.3
P2	20	621.7	718.6	15.6
P3	30	537.4	628.7	17.0
P4	40	432.8	539.4	24.6
P5	50	319.4	441.2	38.1
P6	60	219.8	347.8	58.2
P7	70	149.3	258.3	73.0
P8	80	98.7	174.2	76.5
P9	90	67.2	108.7	61.8
P10	100	40.4	54.8	35.6

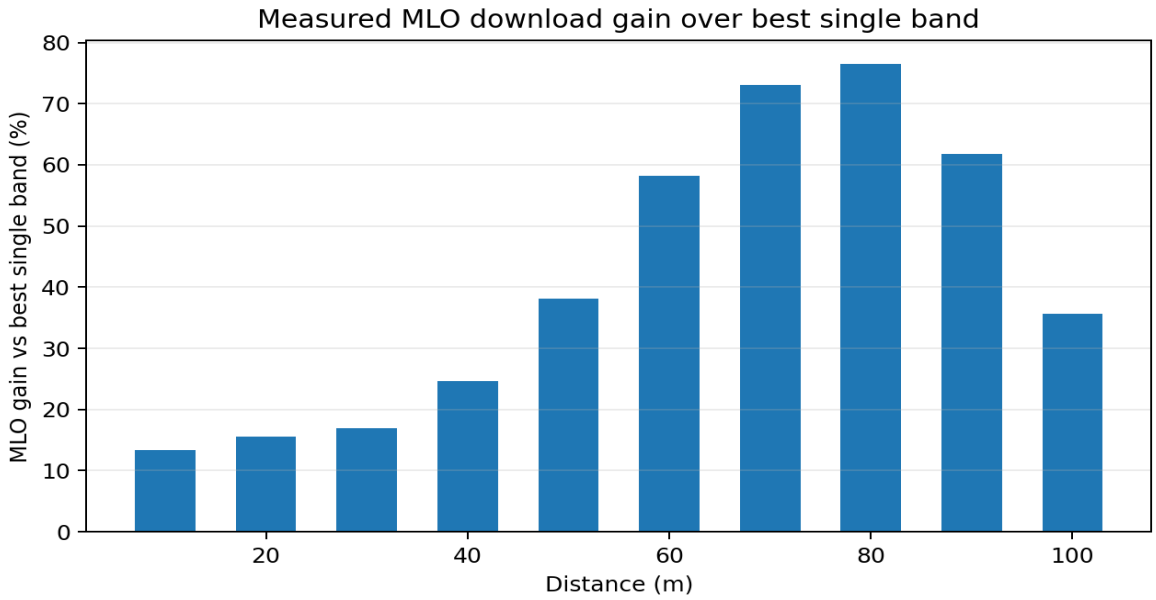


Figure 5. Calculated MLO download gain relative to the best single-band value at each point.

In the measured series, MLO download throughput exceeds the best single-band value at each point. The largest calculated gain occurs at 80 m (approximately 76.5%). This is a calculated comparison of the recorded throughput values, not proof that MLO alone caused the full difference: channel conditions, link scheduling, and test variability may also contribute.

6.6 RSSI Decay

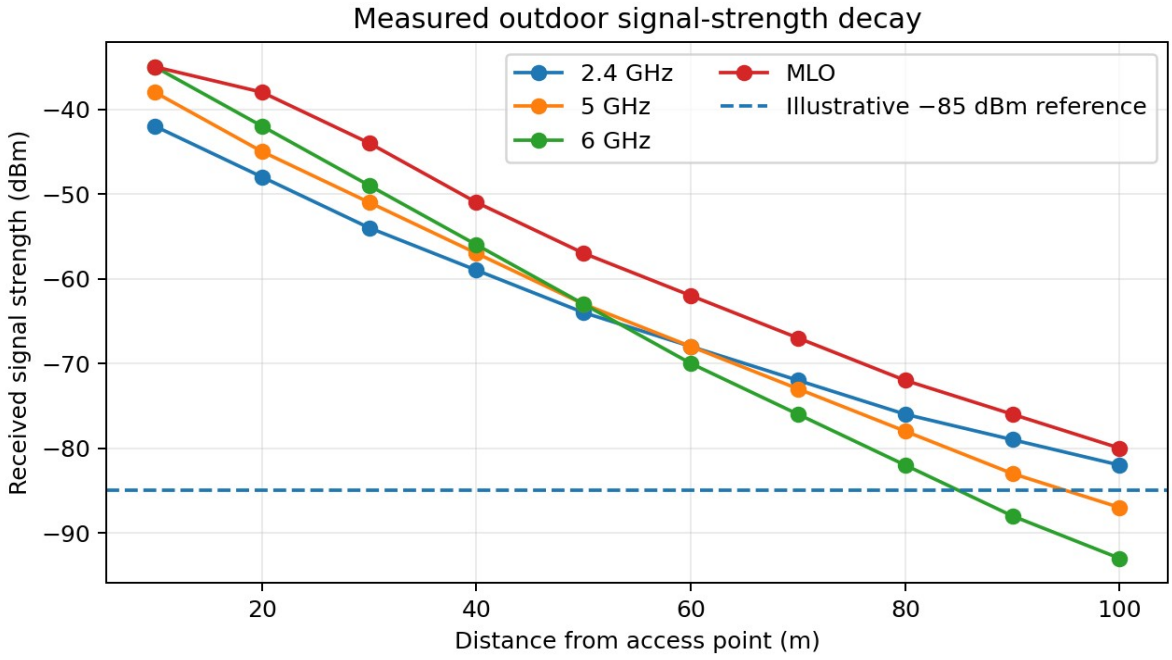


Figure 6. Measured RSSI decline with distance; the dashed line is an illustrative reference only.

RSSI becomes more negative as distance increases across the single-band series. The 6 GHz line falls below the illustrative −85 dBm reference between 80 m and 90 m and reaches −93 dBm at 100 m. The 5 GHz series reaches −87 dBm at 100 m, while 2.4 GHz remains at −82 dBm. The −85 dBm line is a visual reference, not a

universal cutoff: actual usability depends on the device, noise, channel width, modulation, and application requirements.

6.7 Indoor and Outdoor Comparison

Dimension	Part A — Indoor	Part B — Outdoor measurements	Interpretation
Primary metrics	RSSI and SNR	RSSI, SNR, upload/download throughput	The phases use different metric sets by design.
Environment	Rooms, walls, floors	Open field, increasing distance	Indoor obstacles and outdoor distance are distinct factors.
Strongest short-range case	6 GHz baseline: −22 dBm / 48 dB SNR	6 GHz at 10 m: 698.3 Mbps download	Both datasets show strong short-range 6 GHz conditions.
Difficult condition	Floor 3, 6 GHz: −84 dBm / SNR below 1 dB	100 m, 6 GHz: −93 dBm / −3 dB SNR / no connection	High-frequency link quality can deteriorate strongly.
Coverage pattern	All bands weakened by wall/floor obstruction	Lower band remains connected in supplied far point	Band choice should balance capacity and coverage.
Evidence status	Values recorded in project material; raw exports not supplied here	Measured results collected at William Green Domain	Do not claim equivalently validated evidence.

6.8 Propagation Model Considerations

FSPL is useful as a baseline for an unobstructed path, but it does not capture all outdoor or indoor mechanisms. A log-distance model can represent empirical attenuation with distance if its path-loss exponent is estimated from actual observations. ITU-R P.1238-9 is focused on indoor propagation and is most appropriate when the environment and frequency assumptions match its scope [3]. The measurements provide an empirical description of the tested site, but fitting and validating a reliable site-specific propagation model would require complete environmental metadata and a suitable statistical analysis of the observed values.

7. Discussion

7.1 Capacity and Coverage Trade-off

Across the supplied measured outdoor series, the single-band ranking changes with distance. 6 GHz provides the highest download value at short range but degrades most severely at the far endpoint. 5 GHz offers strong short-range performance but also declines substantially. 2.4 GHz has lower short-range throughput yet retains a non-zero value at 100 m. This supports a general planning principle: the band that maximises peak throughput is not necessarily the band that maximises coverage.

7.2 MLO as a Candidate Operating Mode

The measured MLO series exceeds the strongest single-band download value at each point, with gains varying across distance. In deployment, MLO performance remains dependent on the access point/client pair and configuration. Traffic distribution, link scheduling, band availability, client movement, and radio conditions may affect the outcome. The observed gain applies to this measurement campaign and does not prove that MLO will improve performance in every Wi-Fi 7 deployment.

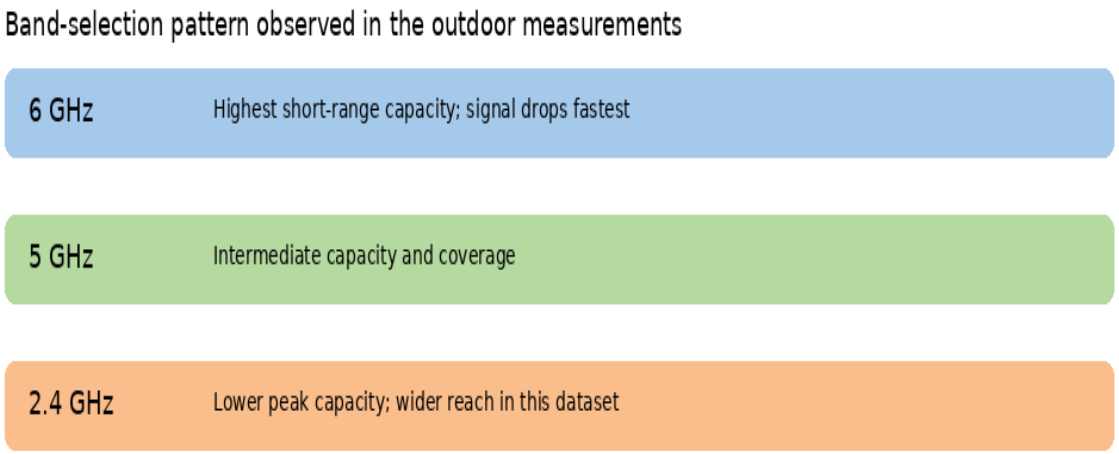


Figure 7. Conceptual interpretation of band-selection tendencies in the supplied MLO series.

7.3 Indoor Deployment Implications

Indoor planning should account for wall materials, the number of obstructions, floor slabs, access-point placement, and the intended client locations. A single access point on Floor 1 may not provide reliable 6 GHz coverage on Floor 3 in the scenario represented here. Possible mitigations include additional access points, wired backhaul, or revised placement. The appropriate solution should be verified through a site survey rather than selected solely from band-level generalisations.

7.4 Outdoor Access-Point Placement

The project record suggests a central-field placement as a candidate configuration for a pitch-wide service area. A central placement can reduce the maximum distance to the farthest points compared with an edge placement, but the optimum position depends on actual pitch dimensions, antenna pattern, mounting height, legal radio limits, client density, and required minimum throughput. The supplied measured dataset is not sufficient to prove an optimum placement. A field survey should compare centre and edge positions with the same client, height, and traffic profile.

7.5 Industry Relevance

For sports and outdoor facilities, link design may support event operations, staff communications, media workflows, and connected services. A robust design should define a service target first—for example, a minimum download rate, coverage boundary, and acceptable packet loss—then validate the design against that target. Maximum peak throughput alone is not a complete acceptance criterion.

7.6 Limitations

- The outdoor measurements were completed at William Green Domain. The main evidence limitation in this report is that the original raw iPerf3 logs and NetSpot exports are not embedded alongside the summary tables, so an independent reviewer cannot reproduce every reading from this document alone.
- The original NetSpot heatmap exports and complete raw indoor measurement logs were not included among the available source files for this report.
- The outdoor dataset does not include full channel width, channel utilisation, transmit power, antenna orientation, weather, client movement, or interference metadata.

- The MLO primary-band sequence is supplied in the project record, but no device logs were provided to verify the link-selection mechanism.
- A limited number of scenario values cannot establish universal range limits or propagation rules across different buildings and outdoor sites.
- The report's charts are generated from the supplied numeric series; they are not photographs or screenshots of actual field instrumentation.

7.7 Validity and Reproducibility

The calculations and plots are reproducible from the numerical series transcribed into the tables. The outdoor values are field measurements, but independent auditability is limited when raw iPerf3 logs, NetSpot exports, timestamps, and full device configuration are not attached to the report. This is an evidence-packaging limitation, not a claim that the outdoor session did not occur. Retaining those files alongside the report allows readers to verify individual readings and reproduce the calculations.

8. Conclusions and Measurement Clarification

8.1 Part A Conclusions

- The supplied indoor baseline records the strongest signal at 6 GHz (−22 dBm, SNR 48 dB) in the same-room scenario.
- Signal strength decreases as the number of intervening walls increases across all three bands.
- The multi-floor values indicate that floor separation can produce a weak or unreliable link, particularly for 6 GHz at Floor 3.
- The results support multi-point indoor surveys and careful access-point placement rather than relying on a single baseline measurement.

8.2 Part B Conclusions

- In the measured outdoor dataset, 6 GHz offers the highest single-band download throughput at 10 m (698.3 Mbps), but the dataset records no connection at 100 m.
- The 2.4 GHz series has lower short-range throughput but remains non-zero at the 100 m endpoint in the measurements.
- The measured MLO series remains non-zero at all ten points and is higher than the best single-band value at each point in the table.
- The calculated MLO gain peaks at approximately 76.5% at 80 m in the measured series.
- These are measured field results from William Green Domain and may be cited as such when the site, test configuration, and measurement conditions are stated. They should not be interpreted as universal performance guarantees.

8.3 Recommendations

- For indoor networks, survey the actual service locations and include wall and floor effects in the design.
- Use 6 GHz where high capacity is needed and the path is favourable; verify coverage beyond the immediate area.
- Consider 5 GHz as an intermediate capacity/coverage option and 2.4 GHz where broader reach is important, subject to congestion and regulatory constraints.
- Evaluate MLO using compatible devices and repeated, controlled trials rather than assuming a fixed performance gain.

- For a sports field, compare central and edge AP positions against defined service targets and measured worst-case points.
- **Traceability.** Preserve the original iPerf3 outputs, NetSpot survey files, timestamps, device settings, and field notes with the final submission. Their inclusion enables independent checking of the completed measurements and the derived plots.

8.4 Measurement Clarification and Interpretation

- **Completed field activity.** The outdoor measurement session was carried out at William Green Domain, Highland Park, using ten points from 10 m to 100 m. The reported outdoor values are the measurements collected during this completed session.
- **Status of the dataset.** The 2.4 GHz, 5 GHz, 6 GHz, and MLO series are measured results, not simulations or estimates. The tables and figures use those recorded values directly.
- **Throughput interpretation.** The reported upload and download figures should be interpreted as the results recorded for the documented test runs. Where repeated-run counts and dispersion statistics are not reported, the table values should not be described as averages or as a statistical estimate of variability.
- **Configuration context.** Radio configuration, client capability, antenna placement, channel conditions, and the MLO implementation can affect results. Conclusions therefore apply to the tested setup at the time of measurement; record exact device and radio settings in the supporting evidence pack where available.
- **Signal evidence.** RSSI and SNR describe received signal conditions and noise margin; they are not interchangeable with application throughput. Original NetSpot exports and screenshots should be retained with the report to support traceability.
- **Indoor/outdoor comparability.** Part A and Part B use different environments and metric emphases. Indoor wall/floor attenuation should not be numerically equated with outdoor distance-related decay because the propagation paths and surrounding conditions differ.
- **MLO comparison.** The reported MLO gain is calculated against the highest single-band download value at the same distance. It is a descriptive comparison of the recorded throughput values and does not, by itself, isolate the causal contribution of MLO from other radio or test conditions.
- **Long-range endpoint.** A recorded zero throughput at 100 m for 6 GHz is interpreted as no connection in that test condition, not as a universal maximum range for the band.
- **Generalisation.** The results characterise the tested William Green Domain setup. Different access points, client adapters, channel widths, transmit powers, antenna positions, interference, and environmental conditions may produce different outcomes.

Final Statement

This combined report analyses Wi-Fi 7 link quality using indoor signal surveys and completed outdoor measurements at William Green Domain. The indoor values highlight the importance of obstructions and floor separation; the outdoor measurements show the changing balance between throughput and coverage and quantify the MLO-versus-single-band comparison for the tested setup. The conclusions are empirical for this measurement campaign, with generalisation limited by site-specific radio conditions and the level of raw evidence included in the report.

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Reference verification note: entries [2], [4], [5], [7], and [10] were assembled from the project handover details. Verify full author lists, titles, publication details, and access dates against the original source documents before submission; incomplete metadata should not be treated as bibliographically verified.

Appendix A — Outdoor Measurement Dataset

The following consolidated table records the measured download throughput series used in Figures 4 and 5. The values were collected at William Green Domain and are not simulated or estimated.

Point	Distance (m)	2.4 GHz DL	5 GHz DL	6 GHz DL	MLO DL (Mbps)
P1	10	348.7	541.6	698.3	791.3
P2	20	321.3	493.2	621.7	718.6
P3	30	289.4	437.1	537.4	628.7
P4	40	251.8	371.4	432.8	539.4
P5	50	214.2	298.6	319.4	441.2
P6	60	172.6	219.8	203.7	347.8
P7	70	134.8	149.3	108.4	258.3
P8	80	98.7	91.7	46.2	174.2
P9	90	67.2	49.1	13.8	108.7
P10	100	40.4	21.3	0	54.8

Appendix B — Indoor Results Tables

B1. Baseline

Band	RSSI (dBm)	SNR (dB)	Channel / note
2.4 GHz	-28	36	Channel 6
5 GHz	-26	42	Channel 36
6 GHz	-22	48	6 GHz channel 5

B2. Wall Penetration

Band	Scenario	RSSI (dBm)	SNR (dB)
2.4 GHz	1 wall / 1.5 m	-38	28
2.4 GHz	2 walls / 5 m	-48	18
2.4 GHz	3 walls / 10 m	-57	10
5 GHz	1 wall	-35	33
5 GHz	2 walls	-47	21
5 GHz	3 walls	-61	8
6 GHz	1 wall	-31	39
6 GHz	2 walls	-48	20
6 GHz	3 walls	-68	2

B3. Multi-floor

Band	Client location	RSSI (dBm)	SNR (dB)	Assessment
2.4 GHz	Floor 2	-52	16	Fair
2.4 GHz	Floor 3	-67	5	Weak
5 GHz	Floor 2	-55	13	Weak
5 GHz	Floor 3	-71	2	Very weak
6 GHz	Floor 2	-62	6	Weak
6 GHz	Floor 3	-84	0.9	No reliable signal

End of report.